

Algorithms

History, State, Behavior, Enemies of OOP

Yegor Bugayenko

Lecture #1 out of 8
80 minutes

The slidedeck was presented by the author in this [YouTube Video](#).

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Pre-Test

History

Original Intent

Object Thinking vs. Algorithms

Enemies of Object Thinking

Post-Test

WARNING!

In the pursuit of academic enlightenment within this course, it is paramount to caution that the doctrines disseminated may present a potentially hazardous venture if employed in real-life software projects. This inherent risk arises from the potential incongruity with the broadly accepted canon of object-oriented programming and recognized best programming practices. If one remains resolute in their decision to adapt their coding methodologies to align with the principles propagated in this course, it would be prudent to employ a certain degree of foresight. A humorous, yet sincere suggestion, would be to secure **alternate employment** prior to a possible premature termination of one's current professional engagement.

Chapter #1:

Pre-Test

[\[quiz \]](#)<https://github.com/yegor256/quiz>

```
1 public class Parser {
2     private File file;
3     public synchronized void setFile(File f) {
4         file = f;
5     }
6     public synchronized File getFile() {
7         return file;
8     }
9     public String getContent() throws IOException {
10         // Read the content of the file
11         // and return it.
12     }
13     public String getContentWithoutUnicode() throws IOException {
14         // Read the file and filter out symbols
15         // that are not UTF-8 compliant.
16     }
17     public void saveContent(String content) {
18         // Save the "content" to the file.
19     }
20 }
```


Chapter #2: History

[[Sketchpad](#) Objects Simula-67 OOP Smalltalk Stroustrup C++ Languages Features]

Who Started It?



Ivan Sutherland's seminal **Sketchpad** application was an early inspiration for OOP, created between 1961 and 1962 and published in his Sketchpad Thesis in 1963. Any object could become a “master,” and additional instances of the objects were called “occurrences”. Sketchpad's masters share a lot in common with JavaScript's prototypal inheritance. (c) Wikipedia

[Sketchpad [Objects](#) Simula-67 OOP Smalltalk Stroustrup C++ Languages Features]

Who Invented Objects, Classes, and Inheritance?



Simula was developed in 1965 at the Norwegian Computing Center in Oslo, by Ole-Johan Dahl and Kristen Nygaard. Like Sketchpad, Simula featured objects, and eventually introduced classes, class inheritance, subclasses, and virtual methods. (c) Wikipedia

[Sketchpad Objects [Simula-67](#) OOP Smalltalk Stroustrup C++ Languages Features]

Simula-67: Sample Code

```
1 Class Figure;  
2   Virtual: Real Procedure area Is Procedure area;;  
3 Begin  
4 End;  
5 Figure Class Circle (c, r);  
6   Real c, r;  
7 Begin  
8   Real Procedure area;  
9   Begin  
10    area := 3.1415 * r * r;  
11  End;  
12 End;
```

[Sketchpad Objects Simula-67 [OOP](#) Smalltalk Stroustrup C++ Languages Features]

Who Coined the “OOP” Term?



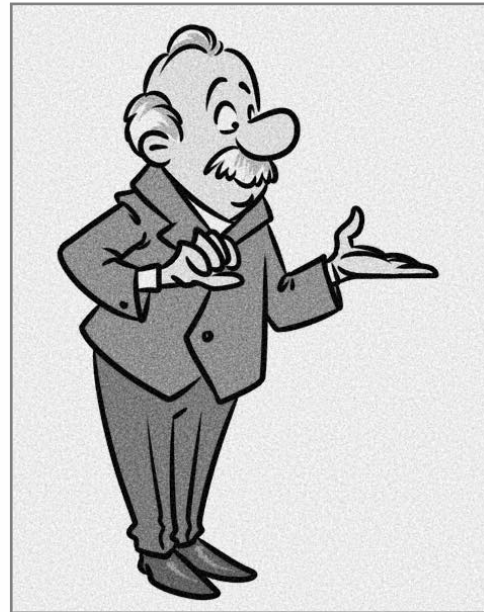
Smalltalk was created in the 1970s at Xerox PARC by Learning Research Group (LRG) scientists, including Alan Kay, Dan Ingalls, Adele Goldberg, Ted Kaehler, Diana Merry, and Scott Wallace. (c) Wikipedia

[Sketchpad Objects Simula-67 OOP [Smalltalk](#) Stroustrup C++ Languages Features]

Smalltalk: Sample Code

```
1 Object subclass: Account [  
2     | balance |  
3     Account class >> new [  
4         | r |  
5         r := super new. r init. ^r  
6     ]  
7     init [ balance := 0 ]  
8 ]  
9 Account extend [  
10     deposit: amount [ balance := balance + amount ]  
11 ]  
12 a := Account new  
13 a deposit: 42
```

[Sketchpad Objects Simula-67 OOP [Smalltalk](#) Stroustrup C++ Languages Features]



“Everyone will be in a favor of OOP. Every manufacturer will promote his products as supporting it. Every manager will pay lip service to it. Every programmer will practice it (differently). And no one will know just what it is.”

— Tim Rentsch. “Object Oriented Programming”. In: *ACM SIGPLAN Notices* 17.9 (1982), pp. 51–57. DOI: [10.1145/947955.947961](https://doi.org/10.1145/947955.947961)

[Sketchpad Objects Simula-67 OOP Smalltalk [Stroustrup](#) C++ Languages Features]

Who Made It All Popular?



C++ was created by Danish computer scientist Bjarne Stroustrup in 1985, by enhancing the C language with Simula-like features. C was chosen because it was general-purpose, fast, portable and widely used.

You may enjoy watching this [one-hour dialog](#) of Dr. Stroustrup and me.

C++: Sample Code

```
1 class Figure {  
2     virtual float area() = 0;  
3 };  
4 class Circle : public Figure {  
5     Circle(float c, float r) : c(c), r(r) {};  
6     float area() { return 3.1415 * r * r; };  
7 private:  
8     float c, r;  
9 };
```

[Sketchpad Objects Simula-67 OOP Smalltalk Stroustrup [C++](#) Languages Features]



OLE LEHRMANN MADSEN

“There are as many definitions of OOP as there are papers and books on the topic.”

— Ole Lehrmann Madsen and Birger Møller-Pedersen. “What Object-Oriented Programming May Be — And What It Does Not Have to Be”. In: *Proceedings of the European Conference on Object-Oriented Programming*. Springer. 1988, pp. 1–20. DOI: [10.1007/3-540-45910-3_1](https://doi.org/10.1007/3-540-45910-3_1)

[Sketchpad Objects Simula-67 OOP Smalltalk Stroustrup [C++](#) Languages Features]



“I made up the term ‘object-oriented,’ and I can tell you I didn’t have C++ in mind.”

— Alan Kay. *The Computer Revolution Hasn’t Happened yet*. 1997. DOI: [10.1007/978-3-322-89884-5_3](https://doi.org/10.1007/978-3-322-89884-5_3)

[Sketchpad Objects Simula-67 OOP Smalltalk Stroustrup [C++](#) Languages Features]

There was an interesting debate between Alan Kay and a few readers of my blog, in the comments section under this blog post: [Alan Kay Was Wrong About Him Being Wrong \[2\]](#).

What Happened Later?

C++ was released in 1985. And then...

Erlang (1986)

Eiffel (1986)

Self (1987)

Perl (1988)

Haskell (1990)

Python (1991)

Lua (1993)

JavaScript (1995)

Ruby (1995)

Java (1995)

PHP3 (1998)

C# (2000)

Go (2009)

Rust (2010)

Swift (2014)

EO (2016)

[Sketchpad Objects Simula-67 OOP Smalltalk Stroustrup C++ [Languages](#) Features]



“There is no uniformity or an agreement on the set of features and mechanisms that belong in an OO language as the paradigm itself is far too general.”

— Oscar Nierstrasz. *A Survey of Object-Oriented Concepts*. Citeseer, 1989

[Sketchpad Objects Simula-67 OOP Smalltalk Stroustrup C++ Languages [Features](#)]

Incomplete List of OOP Features, ... So Far:

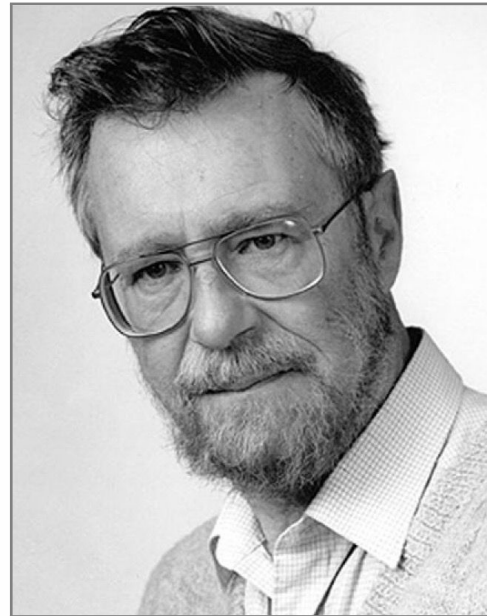
Polymorphism
Nested Objects
Traits
Templates
Generics
Invariants
Classes
NULL
Exceptions
Operators
Methods
Static Blocks
Virtual Tables
Coroutines

Monads
Algebraic Types
Annotations
Interfaces
Constructors
Destructors
Lifetimes
Volatile Variables
Synchronization
Macros
Inheritance
Overloading
Tuple Types
Closures

Access Modifiers
Pattern Matching
Enumerated Types
Namespaces
Modules
Type Aliases
Decorators
Lambda Functions
Type Inference
Properties
Value Types
Multiple Inheritance
Events
Callbacks

NULL Safety
Streams
Buffers
Iterators
Generators
Aspects
Anonymous Objects
Anonymous Functions
Reflection
Type Casting
Lazy Evaluation
Garbage Collection
Immutability

[Sketchpad Objects Simula-67 OOP Smalltalk Stroustrup C++ Languages [Features](#)]



EDSGER DIJKSTRA

“Object oriented programs are offered as alternatives to correct ones... Object-oriented programming is an exceptionally bad idea which could only have originated in California.”

— Bob Crawford. *A Series of Articles About Object Oriented Programming*. TUG LINES, 1989

[Sketchpad Objects Simula-67 OOP Smalltalk Stroustrup C++ Languages [Features](#)]



ALAN KAY

“I made up the term ‘object-oriented,’ and I can tell you I didn’t have C++ in mind.”

— Alan Kay. *The Computer Revolution Hasn’t Happened yet*. 1997. DOI: [10.1007/978-3-322-89884-5_3](https://doi.org/10.1007/978-3-322-89884-5_3)

[Sketchpad Objects Simula-67 OOP Smalltalk Stroustrup C++ Languages [Features](#)]



LINUS TORVALDS

“C++ is a horrible language. . . C++ leads to really, really bad design choices. . . In other words, the only way to do good, efficient, and system-level and portable C++ ends up to limit yourself to all the things that are basically available in C.”

— Esther Schindler. *Linus Torvalds: Why C++ Sucks*.
<https://jttu.net/schindler2007>. [Online; accessed 08-04-2025]. CIO,
2007

[Sketchpad Objects Simula-67 OOP Smalltalk Stroustrup C++ Languages [Features](#)]



JEFF ATWOOD

“OO seems to bring at least as many problems to the table as it solves.”

— Jeff Atwood. *Your Code: OOP or POO?*.
<https://jttu.net/atwood2007>. [Online; accessed 08-04-2025]. 2007

[Sketchpad Objects Simula-67 OOP Smalltalk Stroustrup C++ Languages [Features](#)]



RICH HICKEY

“I think that large object-oriented programs struggle with increasing complexity as you build this large object graph of mutable objects. You know, trying to understand and keep in your mind what will happen when you call a method and what will the side effects be.”

— Rich Hickey. *SE Radio 158: Rich Hickey on Clojure*.
<https://jttu.net/hickey2010>. [Online; accessed 08-04-2025]. 2010

The complexity of object-oriented code
remains its primary drawback

[Sketchpad Objects Simula-67 OOP Smalltalk Stroustrup C++ Languages [Features](#)]



“Reading an OO code you can’t see the big picture and it is often impossible to review all the small functions that call the one function that you modified.”

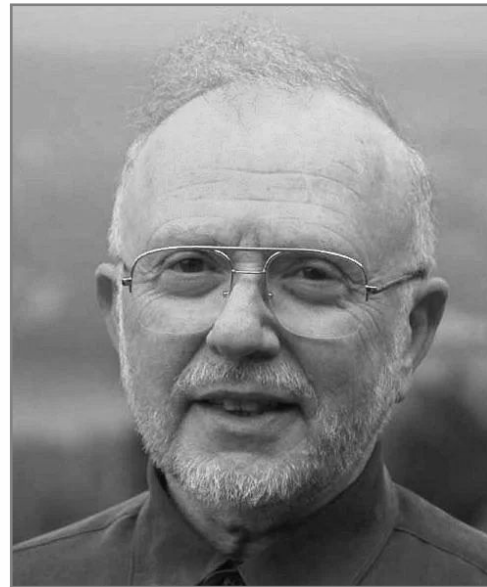
— Asaf Shelly. *Flaws of Object Oriented Modeling*.
<https://jttu.net/shelly2015flaws>. [Online; accessed 15-03-2016]. 2015

Thus, we don't know anymore what exactly is object-oriented programming, and whether it helps us write better code.

You can find more quotes in this blog post of mine: [What's Wrong With Object-Oriented Programming?](#) [3]

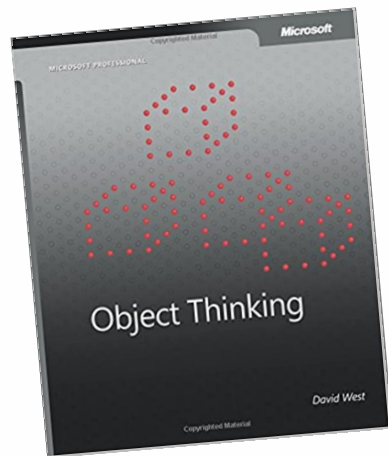
Chapter #3:

Original Intent



“It is almost always incorrect to begin the decomposition of a system into modules on the basis of a flowchart. We propose instead that one begins with a list of difficult design decisions or design decisions which are likely to change. Each module is then designed to hide such a decision from the others.”

— David Lorge Parnas. “On the Criteria to Be Used in Decomposing Systems into Modules”. In: *Communications of the ACM* 15.12 (1972), pp. 1053–1058. DOI: [10.1145/361598.361623](https://doi.org/10.1145/361598.361623)



“The contemporary mainstream understanding of objects (which is not behavioral) is but a pale shadow of the original idea and antithetical to the original intent.”

— David West. *Object Thinking*. Pearson Education, 2004. DOI: [10.5555/984130](https://doi.org/10.5555/984130)

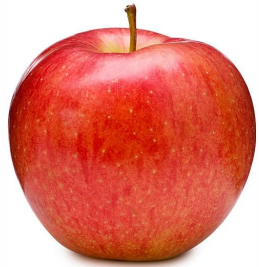
You may enjoy watching our conversation with Dr. David West, video-recorded and published on YouTube: [part I](#) and [part II](#).

A system is a composition of objects that are abstractions, which hide data and expose behavior*

* This is how I understand the original intent.

[Abstraction Rectangle Levels Rectangle Rectangle Function State FigureUtils Composition]

1) What Is an “Abstraction”?



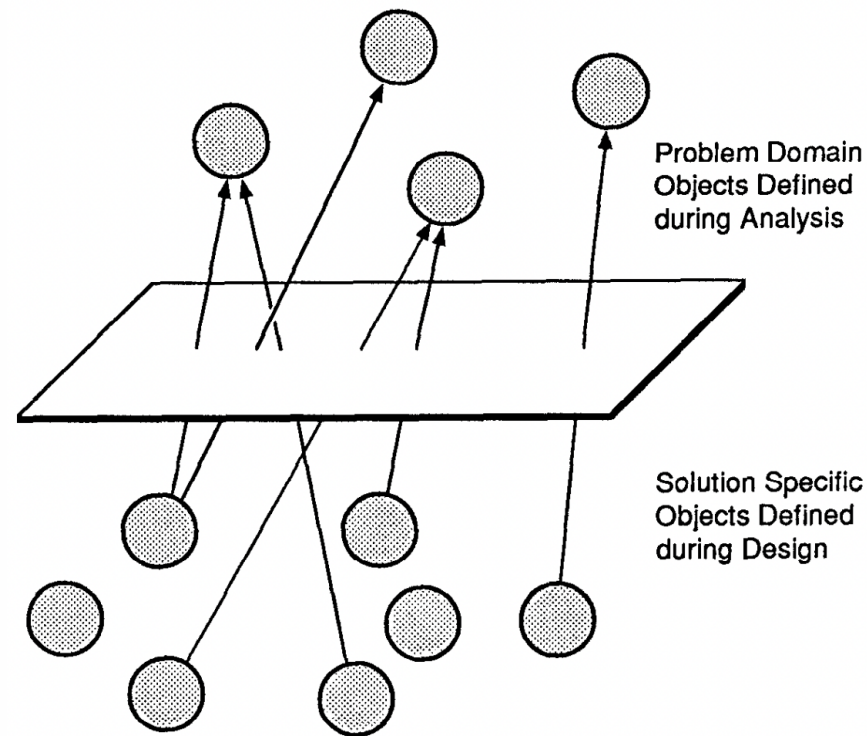
- Color: red
- Weight: 120g
- Price: \$0.99



```
1 var file = {  
2   path: '/tmp/data.txt',  
3   read: function() { ... },  
4   write: function(txt) { ... }  
5 }
```

We deal with an abstraction as if it was a real thing, but eliminating unnecessary details. We do `file.read()` instead of “open file handler for data.txt, read byte by byte, store in byte buffer, wait for the end of file, and return the result.”

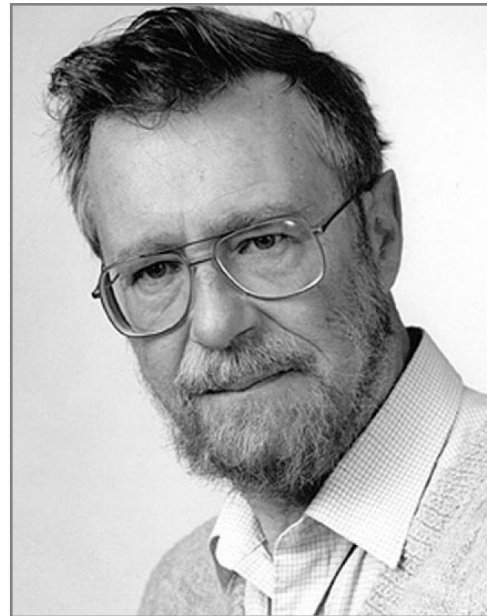
[Abstraction Rectangle Levels Rectangle Rectangle Function State FigureUtils Composition]



“Object-oriented design is first concerned with entities—things. These things may be tangible objects such as traffic lights, chairs, or airplanes. The entities may be abstract concepts such as roles, interactions, or incidents. From a design perspective, objects model the entities in the application domain.”

Source: Tim Korson and John D. McGregor.
“Understanding Object-Oriented: A Unifying Paradigm”.
In: *Communications of the ACM* 33.9 (1990),
pp. 40–60. DOI: [10.1145/83880.84459](https://doi.org/10.1145/83880.84459)

[Abstraction Rectangle Levels Rectangle Rectangle Function State FigureUtils Composition]



“The effective exploitation of his powers of abstraction must be regarded as one of the most vital activities of a competent programmer... By suitable application of our powers of abstraction, the intellectual effort required to conceive or to understand a program need not grow more than proportional to program length.”

— Edsger W. Dijkstra. “The Humble Programmer”. In: *Communications of the ACM* 15.10 (1972), pp. 859–866. DOI: 10.1145/355604.361591

[Abstraction [Rectangle](#) Levels Rectangle Rectangle Function State FigureUtils Composition]

How Many Abstractions Are Needed?

```
1 int area(x1, y1, x2, y2) {  
2     int w = x2 - x1;  
3     if (w < 0) { w = w * -1; }  
4     int h = y2 - y1;  
5     if (h < 0) { h = h * -1; }  
6     return w * h;  
7 }
```

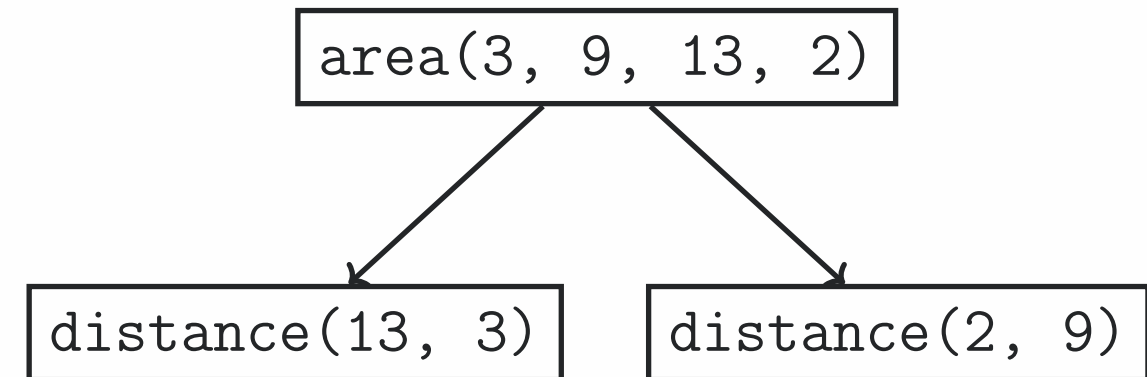
```
1 int distance(left, right) {  
2     int d = right - left;  
3     if (d < 0) { d = d * -1; }  
4     return d;  
5 }  
6  
7 int area(x1, y1, x2, y2) {  
8     return distance(x2, x1)  
9         * distance(y2, y1);  
10 }
```

There are two abstractions at the right snippet (“area” and “distance”), while only one abstraction at the left one (just “area”).

[Abstraction Rectangle [Levels](#) Rectangle Rectangle Function State FigureUtils Composition]

Levels of Abstraction

```
1 int distance(left, right) {  
2   int d = right - left;  
3   if (d < 0) { d = d * -1; }  
4   return d;  
5 }  
6  
7 int area(x1, y1, x2, y2) {  
8   return distance(x2, x1)  
9     * distance(y2, y1);  
10 }
```



Higher level abstractions must not know and/or rely on semantics of lower level abstractions.

2) What Is “Data Hiding”?

```
1 f = new File("/tmp/data.txt");  
2 // The data escapes the object! :(  
3 p = f.getPath();  
4 FileUtils.deleteFile(p);
```

```
1 f = new File("/tmp/data.txt");  
2 // The boolean data escapes too :)  
3 done = f.delete();  
4 assert(done);
```

Obviously, some data must escape your objects.

[Abstraction Rectangle Levels Rectangle [Rectangle](#) Function State FigureUtils Composition]

3) What Is “Behavior Exposing”?

This is so called “anemic” object:

```
1 var user = {  
2   login: 'jeff',  
3   password: 'swordfish',  
4   age: 32  
5 }  
6 function print(u) {  
7   console.log('Hello, ${u.login},  
8     you are ${u.age} today!');  
9 }  
10 print(user);
```

This object is “alive”:

```
1 var user = {  
2   login: 'jeff',  
3   password: 'swordfish',  
4   age: 32,  
5   print: function() {  
6     console.log('Hello, ${this.  
7       login},  
8       you are ${this.age} today!');  
9   }  
10 }  
11 user.print();
```

An Object as a Function

```
1 int distance(left, right) {  
2     int d = right - left;  
3     if (d < 0) { d = d * -1; }  
4     return d; }  
5 int area(x1, y1, x2, y2) {  
6     return distance(x2, x1)  
7         * distance(y2, y1); }
```

```
1 class Distance {  
2     private int r; private int l;  
3     Distance(a, b) { l = a; r = b; }  
4     int value() {  
5         int d = r - l;  
6         if (d < 0) { d = d * -1; }  
7         return d; } }  
8 int area(x1, y1, x2, y2) {  
9     return new Distance(x2, x1).value  
10        ()  
        * new Distance(y2, y1).value();  
    } }
```

[Abstraction Rectangle Levels Rectangle Rectangle [Function](#) State FigureUtils Composition]

The Java object `Distance` on the right snippet is semantically equivalent to the C function `distance()` on the left one.

[Abstraction Rectangle Levels Rectangle Rectangle Function [State](#) FigureUtils Composition]

Identity, State, Behavior

```
1 class Circle {  
2     private float radius;  
3     Circle(float r) {  
4         radius = r; }  
5     float getRadius() {  
6         return radius; }  
7     void setRadius(float r) {  
8         radius = r; }  
9     float area() {  
10         return 3.14 * radius * radius;  
11     }  
}
```

```
1 // Identity:  
2 c1 = new Circle(42.0);  
3 c2 = new Circle(42.0);  
4 c1 != c2;  
5 // State:  
6 c1 = new Circle(42.0);  
7 c2 = new Circle(42.0);  
8 c1.getRadius() == c2.getRadius();  
9 // Behavior:  
10 c1 = new Circle(42.0);  
11 c2 = new Circle(-42.0);  
12 c1.area() == c2.area();
```

[Abstraction Rectangle Levels Rectangle Rectangle Function State [FigureUtils](#) Composition]

State vs. Behavior

```
1 class Circle {
2     private float r;
3     void setR(float r) { this.r = r;
4         }
5     float getR() { return this.r; }
6 }
7 class FigureUtils {
8     static float area(Circle c) {
9         return 3.14 * c.getR() * c.getR
10        ();
11    }
12 }
13 Circle c = new Circle();
14 c.setR(42.0);
15 float s = FigureUtils.area(c);
```

```
1 class Circle {
2     private float r;
3     Circle(float r) { this.r = r; }
4     float area() {
5         return 3.14 * this.r * this.r;
6     }
7 }
8 Circle c = new Circle(42.0);
9 float s = c.area();
```

How to decide what is state and what is behavior?

[Abstraction Rectangle Levels Rectangle Rectangle Function State [FigureUtils](#) Composition]

4) What Is “Composition”?

```
1 canvas = new Canvas();  
2 canvas.addCircle(new Circle(42));  
3 canvas.draw();
```

```
1 canvas = new Canvas();  
2 circle = new Circle(42);  
3 circle.drawOn(canvas);
```

What is composition? What is the “right” composition?

Chapter #4:

Object Thinking vs. Algorithms

[[While](#) Buffer Loop Loop Composition]

While-Do Loop

```
1 buffer = []
2 while true
3   c = STDIN.readchar
4   break if c == "\n"
5   if buffer.length > 3
6     STDOUT.puts buffer.join
7     buffer = []
8   end
9   buffer << c
10 end
```

```
1 $ echo 'Hello, world!' | ruby a.rb
2 Hell
3 o, w
4 orld
```

[While [Buffer](#) Loop Loop Composition]

Buffer Abstraction

```
1 buffer = []
2 while true
3   c = STDIN.readchar
4   break if c == "\n"
5   if buffer.length > 3
6     STDOUT.puts buffer.join
7     buffer = []
8   end
9   buffer << c
10 end
```

```
1 class Buffer
2   def initialize; @data = []; end
3   def push(c)
4     if @data.length > 3
5       STDOUT.puts @data.join
6       @data = []
7     end
8     @data << c
9   end
10 end
11 buffer = Buffer.new
12 while true
13   c = STDIN.readchar
14   break if c == "\n"
15   buffer.push c
16 end
```

[While Buffer [Loop](#) Loop Composition]

Loop Abstraction

```
1 class Buffer
2   def initialize; @data = []; end
3   def push(c)
4     if @data.length > 3
5       STDOUT.puts @data.join
6       @data = []
7     end
8     @data << c
9   end
10 end
11 buffer = Buffer.new
12 while true
13   c = STDIN.readchar
14   break if c == "\n"
15   buffer.push c
16 end
```

```
1 class Buffer
2   # the same
3 end
4 class Pull
5   def initialize(b); @buf = b; end
6   def again
7     c = STDIN.readchar
8     return false if c == "\n"
9     @buf.push c
10    true
11  end
12 end
13 buffer = Buffer.new
14 pull = Pull.new(buffer)
15 while pull.again; end
```

[While Buffer Loop [Loop](#) Composition]

Loop Abstraction

```
1 class Buffer
2   # the same
3 end
4 class Pull
5   def initialize(b); @buf = b; end
6   def again
7     c = STDIN.readchar
8     return false if c == "\n"
9     @buf.push c
10    true
11  end
12 end
13 buffer = Buffer.new
14 pull = Pull.new(buffer)
15 while pull.again; end
```

```
1 class Buffer
2   # the same
3 end
4 class Pull
5   # the same
6 end
7 class Pulls
8   def initialize(p); @pull = p; end
9   def fetch
10    while @pull.again; end
11  end
12 end
13 Pulls.new(Pull.new(Buffer.new)).fetch
```

[While Buffer Loop Loop [Composition](#)]

Object Composition

```
1 class Buffer
2   def initialize; @data = []; end
3   def push(c)
4     if @data.length > 3
5       STDOUT.puts @data.join
6       @data = []
7     end
8     @data << c
9   end
10 end
11
12 class Pull
13   def initialize(b); @buf = b; end
14   def again
15     c = STDIN.readchar
16     return false if c == "\n"
17     @buf.push c
```

```
18     true
19   end
20 end
21
22 class Pulls
23   def initialize(p); @pull = p; end
24   def fetch
25     while @pull.again; end
26   end
27 end
28
29 Pulls.new(
30   Pull.new(
31     Buffer.new
32   )
33 ).fetch
```

Chapter #5:

Enemies of Object Thinking

[[List](#)]

What Makes Us Think as Algorithms

Static Methods

Anemic Objects (Getters)

Mutability (Setters)

Workers (“-er” Suffix)

NULL References

Type Casting (Reflection)

Inheritance

Chapter #6: Post-Test



<https://github.com/yegor256/hangman>

References

- [1] Jeff Atwood. *Your Code: OOP or POO?* <https://jttu.net/atwood2007>. [Online; accessed 08-04-2025]. 2007.
- [2] Yegor Bugayenko. *Alan Kay Was Wrong About Him Being Wrong*. <https://www.yegor256.com/171212.html>. [Online; accessed 08-07-2024]. Dec. 2017.
- [3] Yegor Bugayenko. *What's Wrong With Object-Oriented Programming?* <https://www.yegor256.com/160815.html>. [Online; accessed 08-07-2024]. Aug. 2016.
- [4] Bob Crawford. *A Series of Articles About Object Oriented Programming*. TUG LINES, 1989.
- [5] Edsger W. Dijkstra. "The Humble Programmer". In: *Communications of the ACM* 15.10 (1972), pp. 859–866. DOI: [10.1145/355604.361591](https://doi.org/10.1145/355604.361591).
- [6] Rich Hickey. *SE Radio 158: Rich Hickey on Clojure*. <https://jttu.net/hickey2010>. [Online; accessed 08-04-2025]. 2010.
- [7] Alan Kay. *The Computer Revolution Hasn't Happened yet*. 1997. DOI: [10.1007/978-3-322-89884-5_3](https://doi.org/10.1007/978-3-322-89884-5_3).
- [8] Tim Korson and John D. McGregor. "Understanding Object-Oriented: A Unifying Paradigm". In: *Communications of the ACM* 33.9 (1990), pp. 40–60. DOI: [10.1145/83880.84459](https://doi.org/10.1145/83880.84459).
- [9] Ole Lehrmann Madsen and Birger Møller-Pedersen. "What Object-Oriented Programming May Be — And What It Does Not Have to Be". In: *Proceedings of the European Conference on Object-Oriented Programming*. Springer. 1988, pp. 1–20. DOI: [10.1007/3-540-45910-3_1](https://doi.org/10.1007/3-540-45910-3_1).
- [10] Oscar Nierstrasz. *A Survey of Object-Oriented Concepts*. Citeseer, 1989.
- [11] David Lorge Parnas. "On the Criteria to Be Used in Decomposing Systems into Modules". In: *Communications of the ACM* 15.12 (1972), pp. 1053–1058. DOI: [10.1145/361598.361623](https://doi.org/10.1145/361598.361623).
- [12] Tim Rentsch. "Object Oriented Programming". In: *ACM SIGPLAN Notices* 17.9 (1982), pp. 51–57. DOI: [10.1145/947955.947961](https://doi.org/10.1145/947955.947961).
- [13] Esther Schindler. *Linus Torvalds: Why C++ Sucks*. <https://jttu.net/schindler2007>. [Online; accessed 08-04-2025]. CIO, 2007.
- [14] Asaf Shelly. *Flaws of Object Oriented Modeling*. <https://jttu.net/shelly2015flaws>. [Online; accessed 15-03-2016]. 2015.
- [15] David West. *Object Thinking*. Pearson Education, 2004. DOI: [10.5555/984130](https://doi.org/10.5555/984130).